

Practical no. 7 AI
Naive Bayes' Classifier

Aim: Implement the Naive Bayes' algorithm for classification.
Train a Naive Bayes' model using a given dataset and calculate class probabilities.
Evaluate the accuracy of the model on test data and analyze the results.

Code :

```
ai p7 T117.py - C:\Users\Adarsh\OneDrive\Videos\ai p7 T117.py (3.12.0)
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import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import (
    accuracy_score,
    confusion_matrix,
    classification_report,
    ConfusionMatrixDisplay,
    roc_curve,
    roc_auc_score
)

print("Adarsh Shukla T117")

df = pd.read_csv(r"C:\Users\Adarsh\Downloads\diabetes997.csv")

print("\nDataset:")
print(df.head())

if "Outcome" in df.columns:
    target = "Outcome"
elif "outcome" in df.columns:
    target = "outcome"
elif "Diabetes" in df.columns:
    target = "Diabetes"
else:
    target = df.columns[-1]

X = df.drop(columns=[target])
y = df[target]
```

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```
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elif "Diabetes" in df.columns:
    target = "Diabetes"
else:
    target = df.columns[-1]

X = df.drop(columns=[target])
y = df[target]

X = X.select_dtypes(include=[np.number])
X = X.fillna(X.mean())

if y.dtype == "object":
    y = pd.factorize(y)[0]

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42,
    stratify=y
)

model = GaussianNB()
model.fit(X_train, y_train)

y_pred = model.predict(X_test)
y_prob = model.predict_proba(X_test)[: , 1]

accuracy = accuracy_score(y_test, y_pred)

print("\nClass Probabilities:")
for cls, prob in zip(model.classes_, model.class_prior_):
```

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print("\nAccuracy:", round(accuracy * 100, 2), "%")

print("\nConfusion Matrix:")
cm = confusion_matrix(y_test, y_pred)
print(cm)

print("\nClassification Report:")
print(classification_report(y_test, y_pred))

ConfusionMatrixDisplay(
    confusion_matrix=cm,
    display_labels=model.classes_
).plot()

plt.title("Confusion Matrix - Gaussian Naive Bayes")
plt.xlabel("Predicted Class")
plt.ylabel("Actual Class")
plt.show()

plt.figure(figsize=(8, 5))

plt.scatter(
    range(len(y_test)),
    np.array(y_test),
    label="Actual",
    marker="o"
)

plt.scatter(
    range(len(y_pred)),
    y_pred,
```

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```
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print("\nROC AUC:", round(roc_auc, 4))

plt.figure(figsize=(10, 6))

plt.plot(
    fpr,
    tpr,
    linewidth=2,
    label="Gaussian Naive Bayes (AUC = %.2f)" % roc_auc
)

plt.plot(
    [0, 1],
    [0, 1],
    linestyle="--",
    label="Random Classifier"
)

plt.xlim([0.0, 1.0])
plt.ylim([0.0, 1.05])

plt.xticks(np.arange(0, 1.1, 0.1))
plt.yticks(np.arange(0, 1.1, 0.1))

plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve - Gaussian Naive Bayes")

plt.grid()
plt.legend(loc="lower right")
plt.show()
```

Output :

```
"IDLE Shell 3.12.6"
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Python 3.12.6 (tags/v3.12.6:a4a2d2b, Sep  6 2024, 20:11:23) [MSC v.1940 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/Adarsh/OneDrive/Videos/ai p7 T117.py
Adarsh Shukla T117

Dataset:
  Pregnancies  Glucose  BloodPressure  ...  DiabetesPedigreeFunction  Age  Outcome
0             6      148             72  ...                0.627      50         1
1             1       85             66  ...                0.351      31         0
2             8      183             64  ...                0.672      32         1
3             1       89             66  ...                0.167      21         0
4             0      137             40  ...                2.288      33         1

[5 rows x 9 columns]

Class Probabilities:
Class 0 : 0.6515
Class 1 : 0.3485

Accuracy: 70.78 %

Confusion Matrix:
[[74 26]
 [19 35]]

Classification Report:
              precision    recall  f1-score   support

     0       0.80        0.74        0.77        100
     1       0.57        0.65        0.61         54
```

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```
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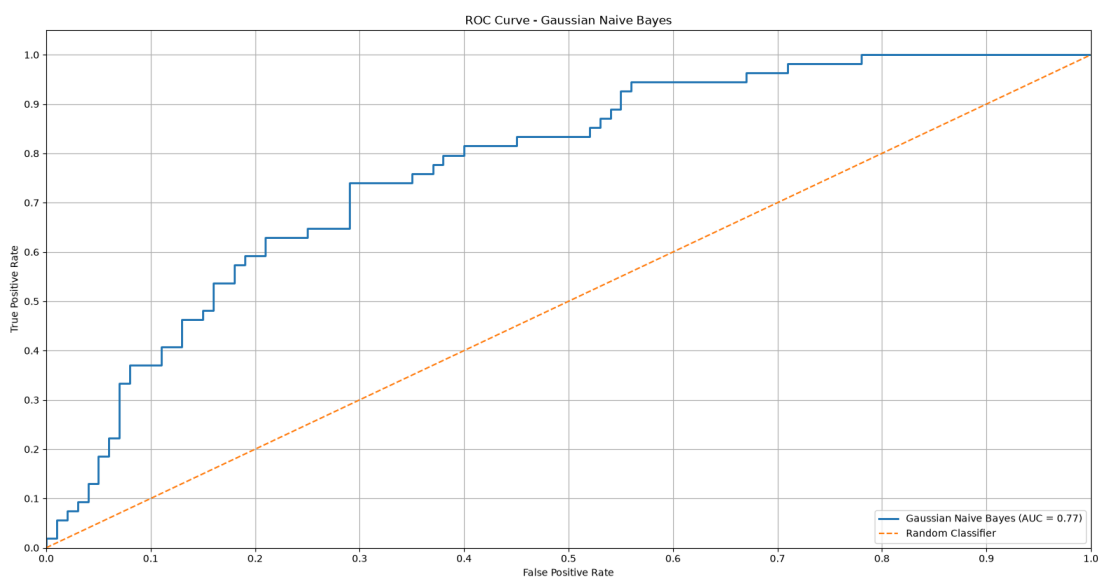
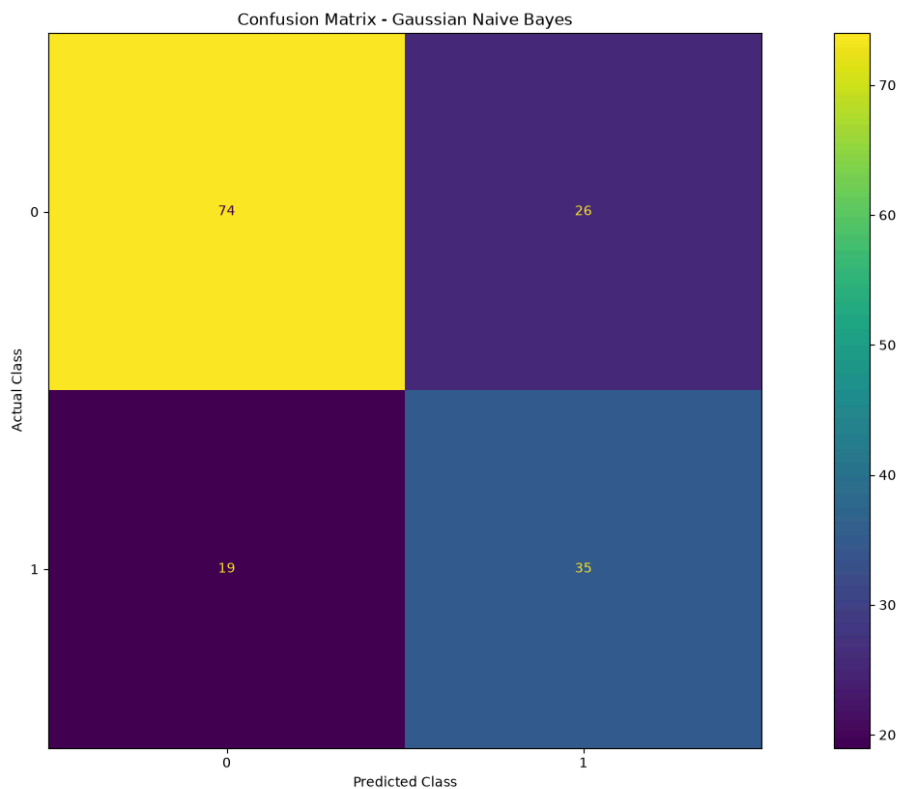
         0       0.80        0.74        0.77        100
         1       0.57        0.65        0.61         54

   accuracy          0.68
  macro avg          0.68
 weighted avg          0.68

ROC Coordinates:
FPR = 0.0 TPR = 0.0
FPR = 0.0 TPR = 0.0185
FPR = 0.01 TPR = 0.0185
FPR = 0.01 TPR = 0.0556
FPR = 0.02 TPR = 0.0556
FPR = 0.02 TPR = 0.0741
FPR = 0.03 TPR = 0.0741
FPR = 0.03 TPR = 0.0926
FPR = 0.04 TPR = 0.0926
FPR = 0.04 TPR = 0.1296
FPR = 0.05 TPR = 0.1296
FPR = 0.05 TPR = 0.1852
FPR = 0.06 TPR = 0.1852
FPR = 0.06 TPR = 0.2222
FPR = 0.07 TPR = 0.2222
FPR = 0.07 TPR = 0.3333
FPR = 0.08 TPR = 0.3333
FPR = 0.08 TPR = 0.3704
FPR = 0.11 TPR = 0.3704
FPR = 0.11 TPR = 0.4074
FPR = 0.13 TPR = 0.4074
FPR = 0.13 TPR = 0.463
FPR = 0.15 TPR = 0.463
FPR = 0.15 TPR = 0.4815
```

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